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PEI 26/27 - Group 2 - 13th October 2026



TreeLens

Say Goodbye to Field Surveys

Recap: Why TreeLens?

Context

Law No. 59/2021 requires municipalities to inventory urban trees, done today tree by tree.

Problem

Process is slow, resource-heavy, and quickly outdated.

Goals

Detect, locate and map every tree automatically, from images and video.

State of the Art

	Manual surveys	i-Tree Eco / Streets	OpenTree Map	iNaturalist	Research pipelines	TreeLens
Automatic detection	❌	❌	❌	❌	✅	✅
Individual tree geolocation	❌	❌	✅	✅	✅	✅
Easy to update and compare	❌	🟨	❌	🟨	❌	✅
No visit to each tree	❌	❌	❌	❌	✅	✅
Ready to use municipal tool	✅	🟨	🟨	❌	❌	✅

Functional Requirements

Approximately **12 functional requirements were collected**, and the most important ones are presented here.



01 The system must **allow the selection of a map zone/the submission of GPS-tagged video**, and detect trees in the resulting images/frames.

02 The system must display the detected trees on an **interactive map**, with individual records (source images, geographic position, characteristics).

03 The system must allow for the confirmation, rejection, or manual correction of a detected tree, and **use these corrections to retrain the detection model**.

04 The system must allow the **export of the georeferenced inventory** to a Municipal Administrator.

05 The system must allow users to **filter the inventory** by zone, species, or validation status.

06 The system must allow a Municipal Administrator to **create survey zones** and **assign them** as tasks to Field Surveyors.

07 The system must maintain a **queue of pending submissions for review**, and allow a Municipal Technician to approve or reject each one before it officially joins the inventory.

08 The system must allow a Municipal Technician to manually **merge/split tree** detections **identified as duplicates**.

Non-Functional Requirements

Approximately **15 non-functional requirements were collected**, and the most important ones are presented here.



01 The system must support **authentication** and **role-based access control**.

02 The system must **comply with the GDPR** regarding the processing of collected personal data.

03 The system must ensure that **each** physical **tree appears only once** in the inventory, even if detected in multiple images.

04 The system must support **graceful degradation**, meaning if the detection pipeline is unavailable, browsing the existing map and inventory must remain functional.

05 The interactive map interface must remain **usable on a mobile device** in outdoor field conditions.

06 All API endpoints must be protected against SQL injection, XSS, and CSRF attacks.

07 All data in transit must be **encrypted** (HTTPS), and stored passwords must use a **strong hashing algorithm**.

08 The average response time for requests, excluding media cataloging, must be 300ms or less.

Actors



Field Surveyor



Municipal Technician



Municipal Administrator

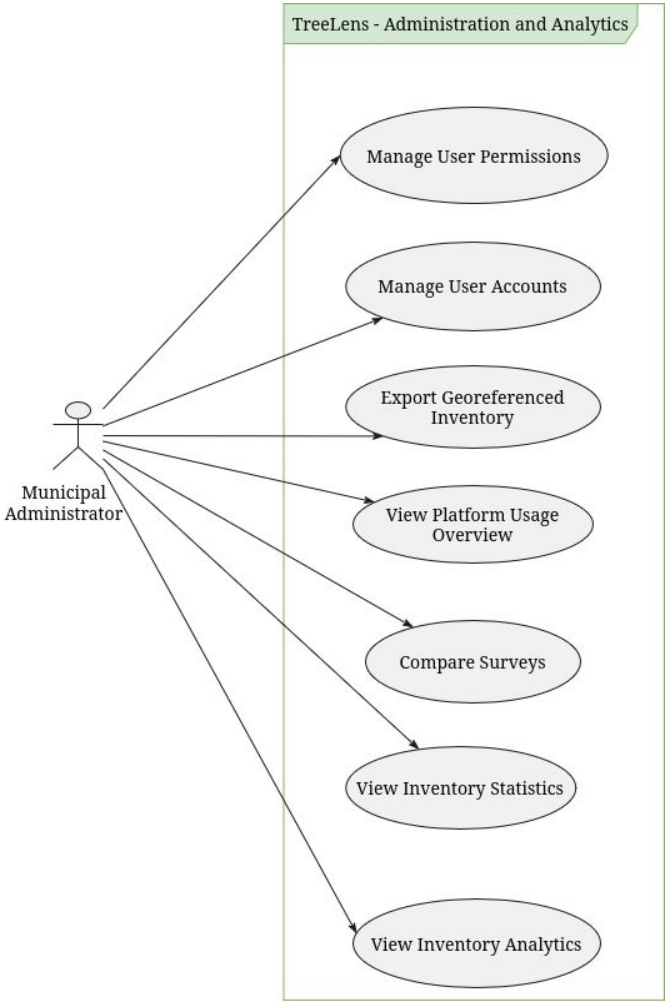


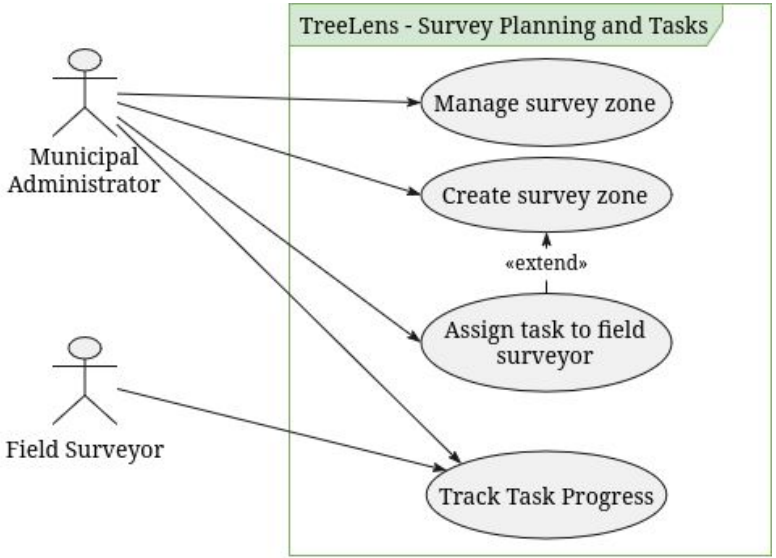
Tree Enthusiast

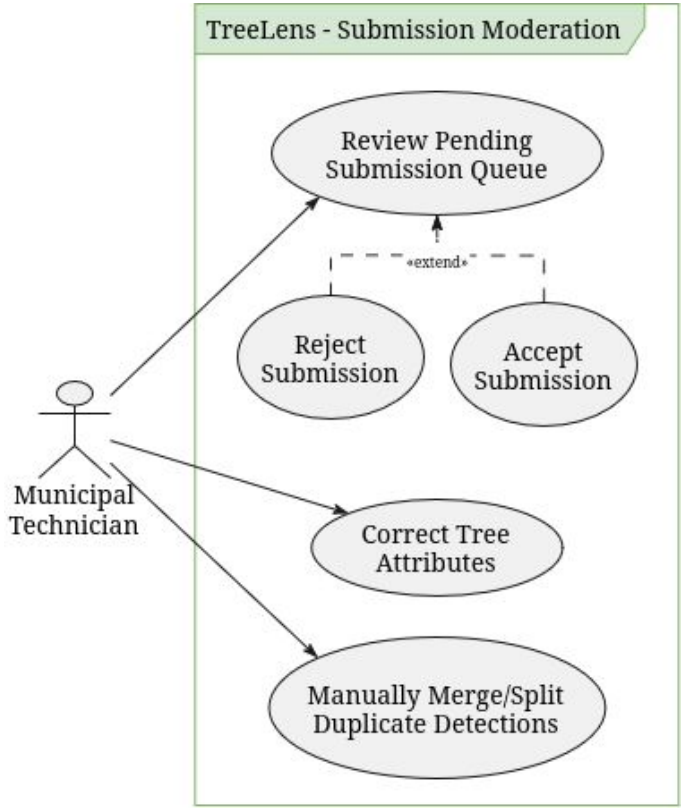
Use Cases

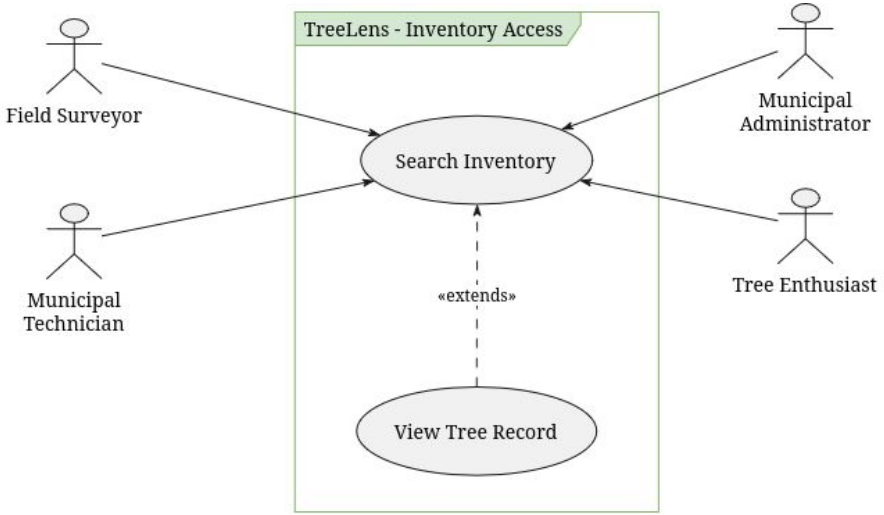
Considering the previously presented actors, the use cases devised for our project follow now.

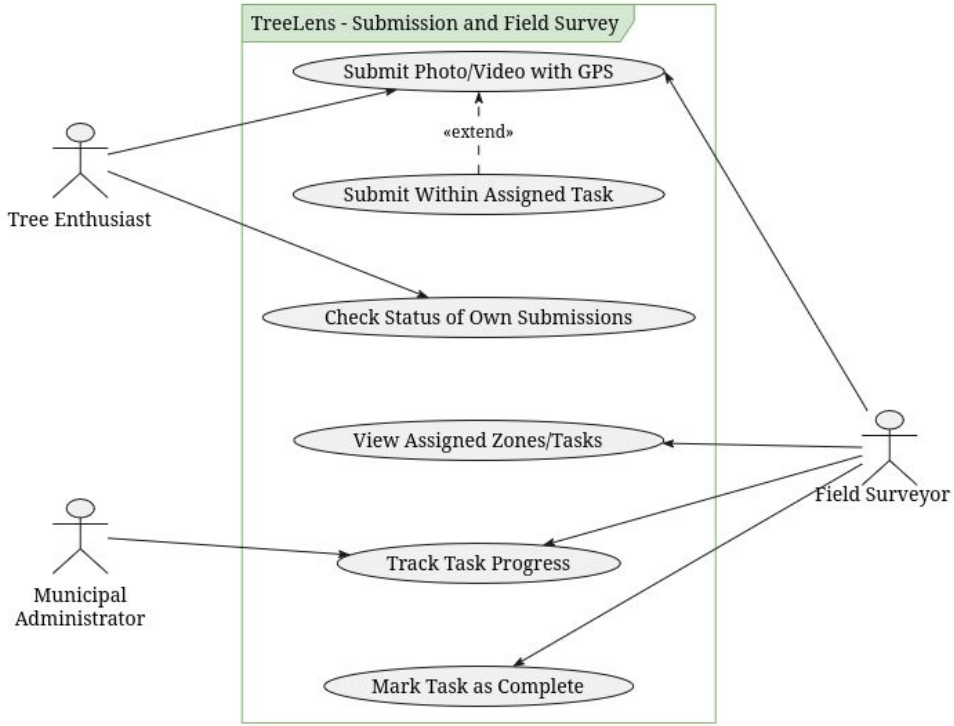



TreeLens - Administration and Analytics
Manage User Permissions
Manage User Accounts
Export Georeferenced Inventory
View Platform Usage Overview
Compare Surveys
View Inventory Statistics
View Inventory Analytics
Municipal Administrator


TreeLens - Survey Planning and Tasks
Municipal Administrator
Manage survey zone
Create survey zone
«extend»
Assign task to field surveyor
Field Surveyor
Track Task Progress


TreeLens - Submission Moderation
Review Pending Submission Queue
«extend»
Reject Submission
Accept Submission
Municipal Technician
Correct Tree Attributes
Manually Merge/Split Duplicate Detections


TreeLens - Inventory Access
Field Surveyor
Municipal Administrator
Search Inventory
Municipal Technician
Tree Enthusiast
«extends»
View Tree Record


TreeLens - Submission and Field Survey
Submit Photo/Video with GPS
«extend»
Submit Within Assigned Task
Tree Enthusiast
Check Status of Own Submissions
View Assigned Zones/Tasks
Field Surveyor
Track Task Progress
Municipal Administrator
Mark Task as Complete

User Stories

Around 32 user stories were grouped together. Here are just a few examples:

01

As a Tree Enthusiast,
I want to submit a photo of a tree
So that I can find out which species it is

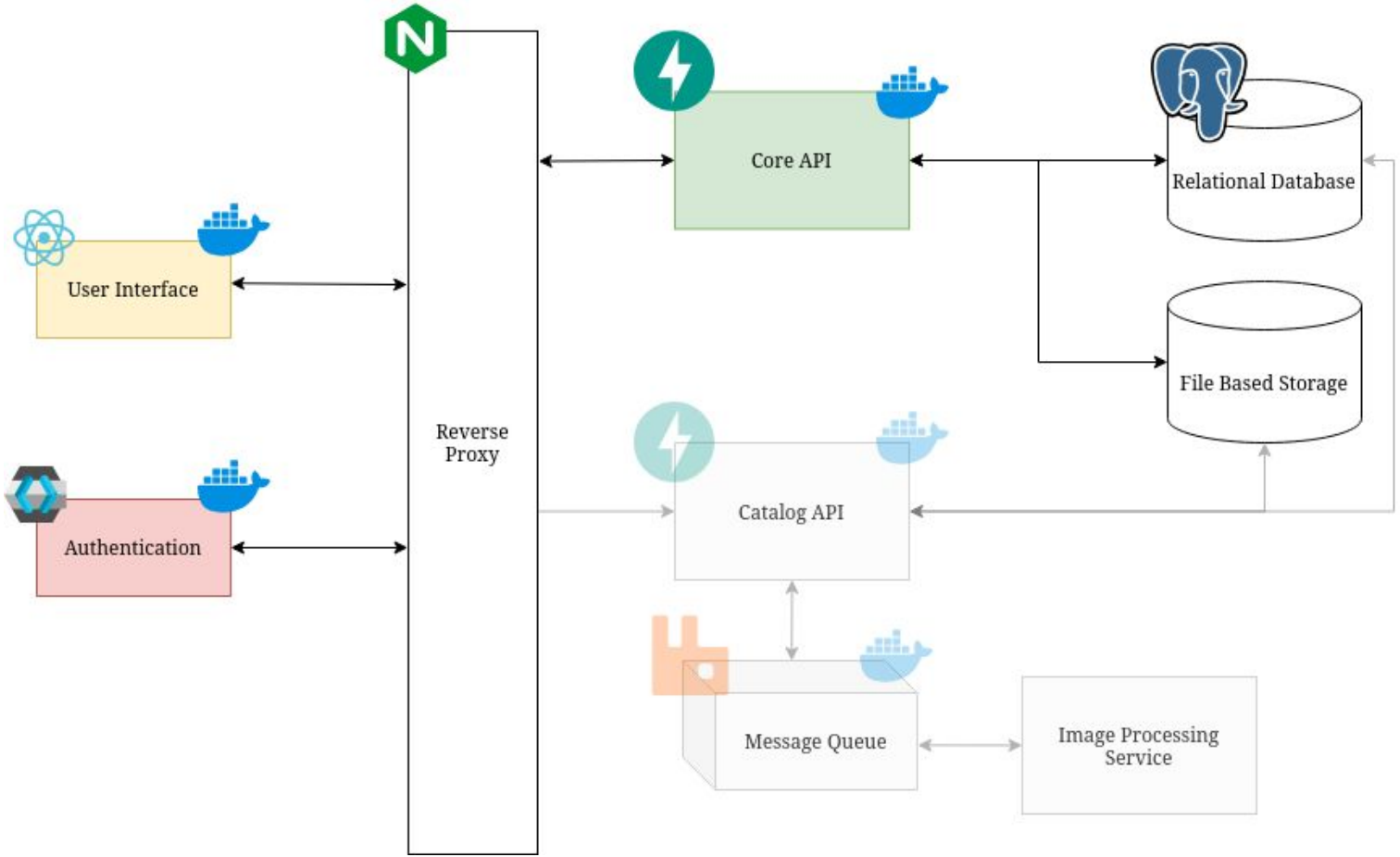
02

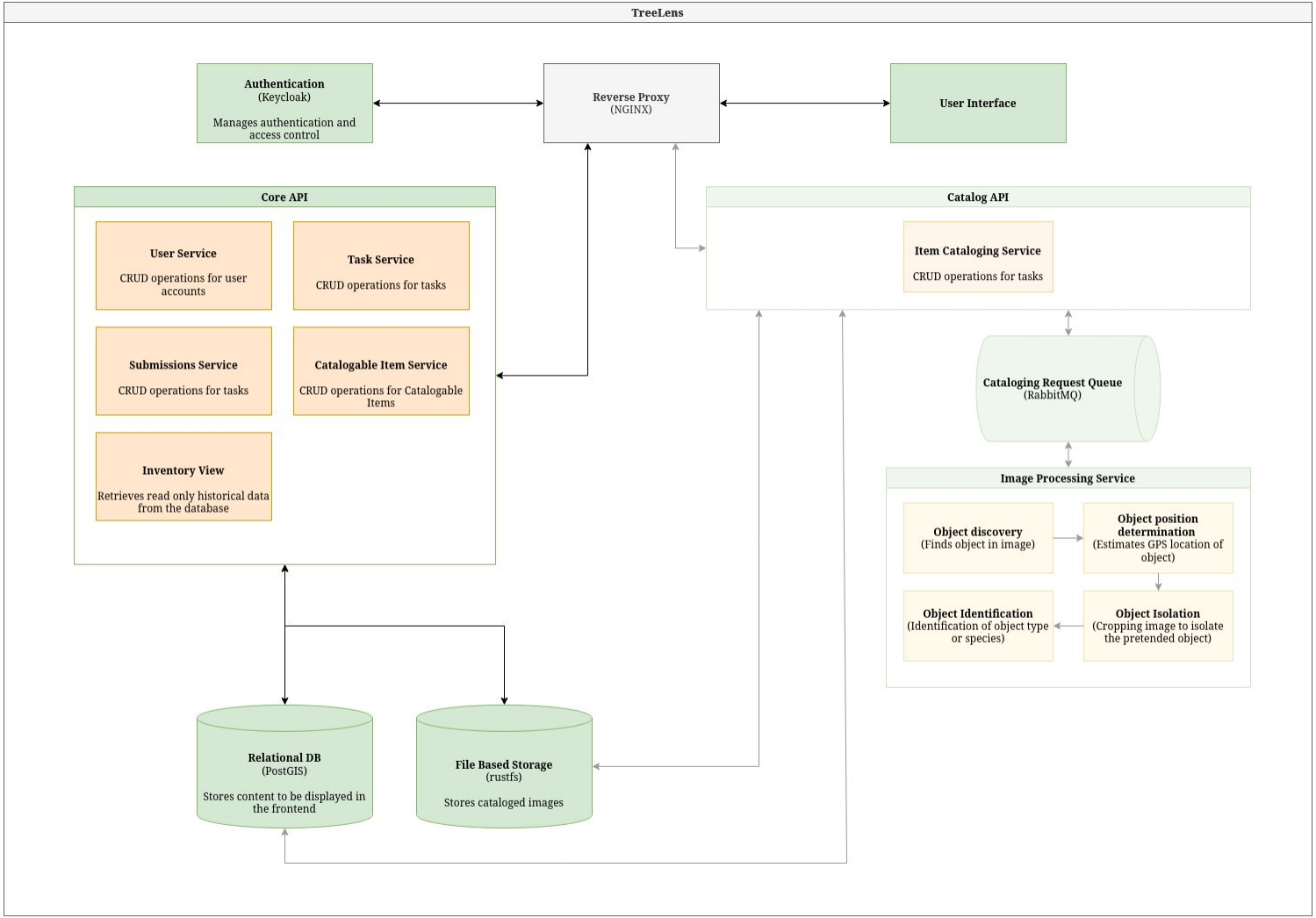
As a Municipal Technician,
I want to approve a pending item
So that reliable data enters the official inventory

03

As a Municipal Administrator,
I want to view statistics about the tree inventory
So that I have the information needed for city management

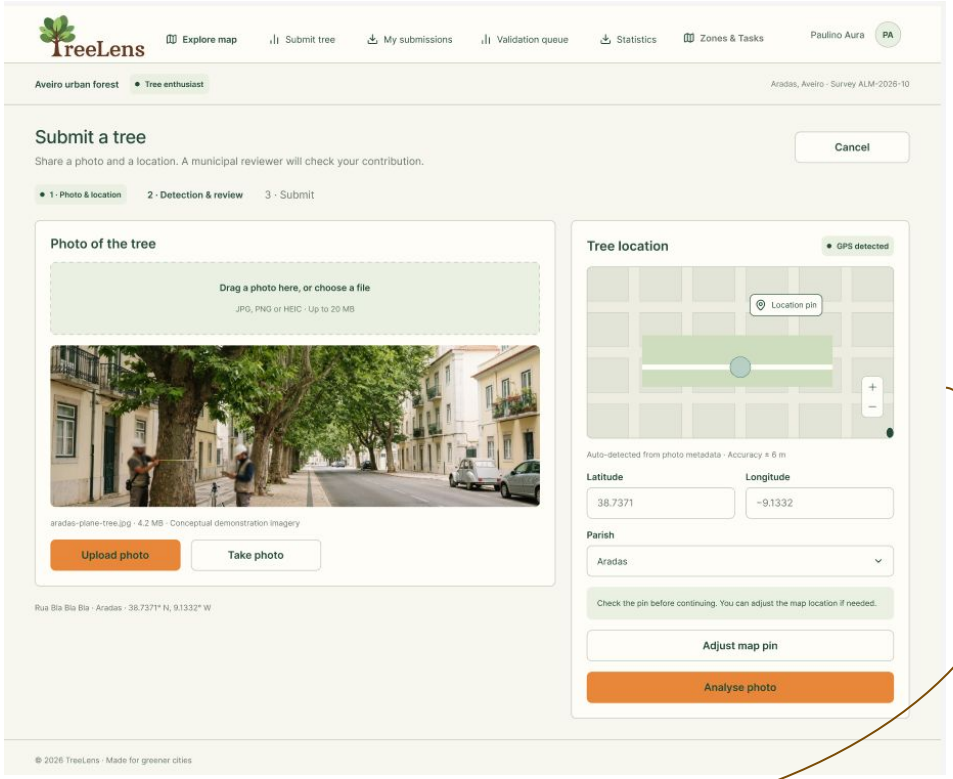
Container-Level Architecture

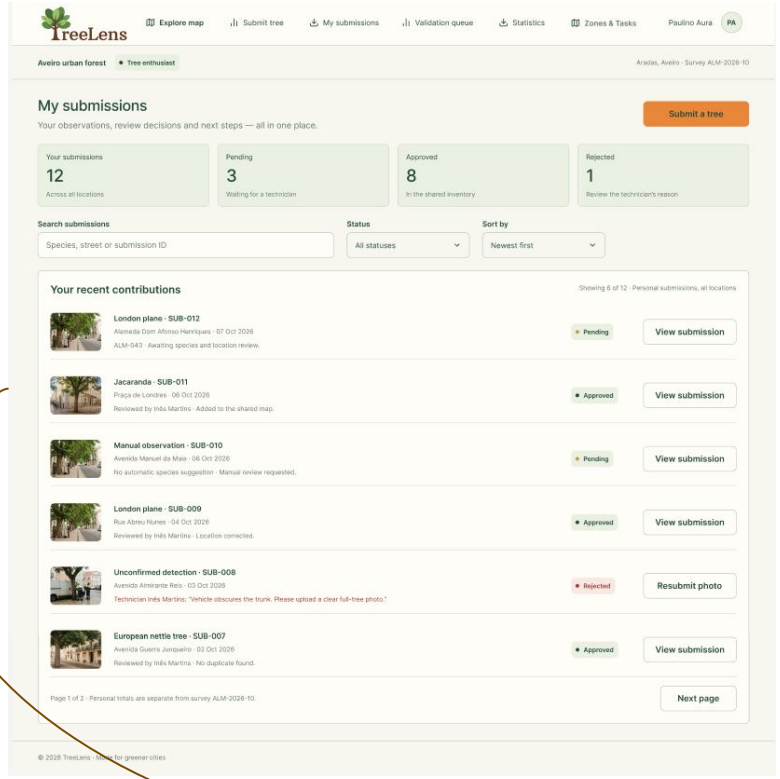




Components -Level Architecture

User Interface

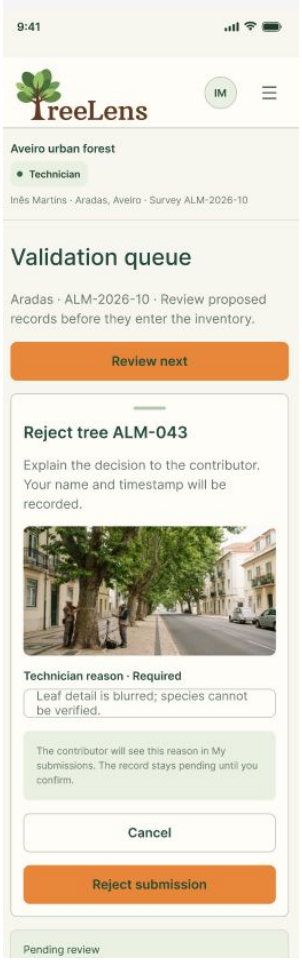


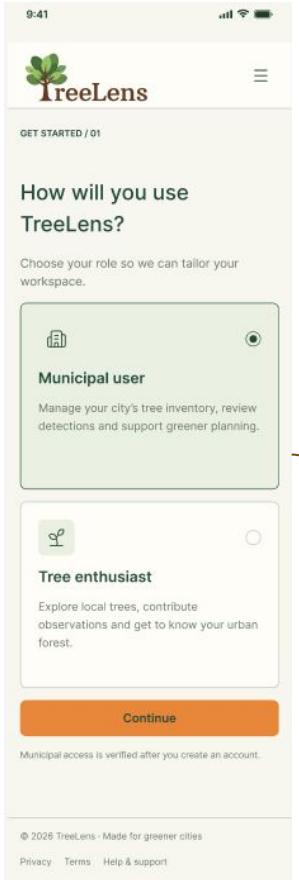


pending queue

upload of an image/video

model's rejection to a detection





role's selection

TreeLens

Also check our complete mockups !



Keep up with our work and check our calendar for the next milestone!